# Rational Number Project

## Fraction Operations and Initial Decimal Ideas
### Lesson 12: Overview

Students review ordering and equivalence and practice adding and subtracting decimals in problem solving contexts.

## Materials

- Student problems A – G cut apart and put in envelopes for each group.
- Poster paper, markers, multiple copies of 10 x 10 grid and Decimal +- board.
- Yellow and orange crayons, pencils or markers.

## Teaching Actions

### Warm Up

Estimate:
.85 + .39 =
Greater than 1 or less then 1?

3.675 + 2.399 =
Greater then 6 or less than 6?

45.06 – 40.56 =
Greater than 5 or less than 5?

### Large Group Introduction

1. In this lesson students have the opportunity to practice their decimal order and equivalence ideas to extend their work with decimal addition and subtraction using symbols.

2. Give each group a set of problems to solve. They can solve in any order they want

## Comments

The decimal lessons have been developed so students construct these indicators of decimal understanding: (1) using precise mathematical language when working with decimals, (2) accurately using models to represent decimals, (3) describing how to compose and decompose decimals through mental images of the models as they ordered decimals, (4) using their understanding of the relative size of decimals to guide their estimation with operations with decimals, and (5) using a model and their ability to compose and decompose decimals to interpret addition and subtraction operations and build meaning for work with symbols.

The rest of the decimal lessons reinforce and extend students' decimal concepts and procedures. You might find that some students may need more time completing lessons 9 – 11. Consider Lesson 12 as an extension for those students who are ready to go beyond Grade 4 and 5 Standards for Decimals.

## Small Group/Partner Work

3. After students have finished half of the problems, assign a particular problem to each group to prepare and present to the whole class. At this point students may not finish the rest of the problems.

4. Provide poster paper, markers, and 10 x 10 grids that can be cut apart and put on posters. Students are to construct a large display of their solution.

## Wrap Up

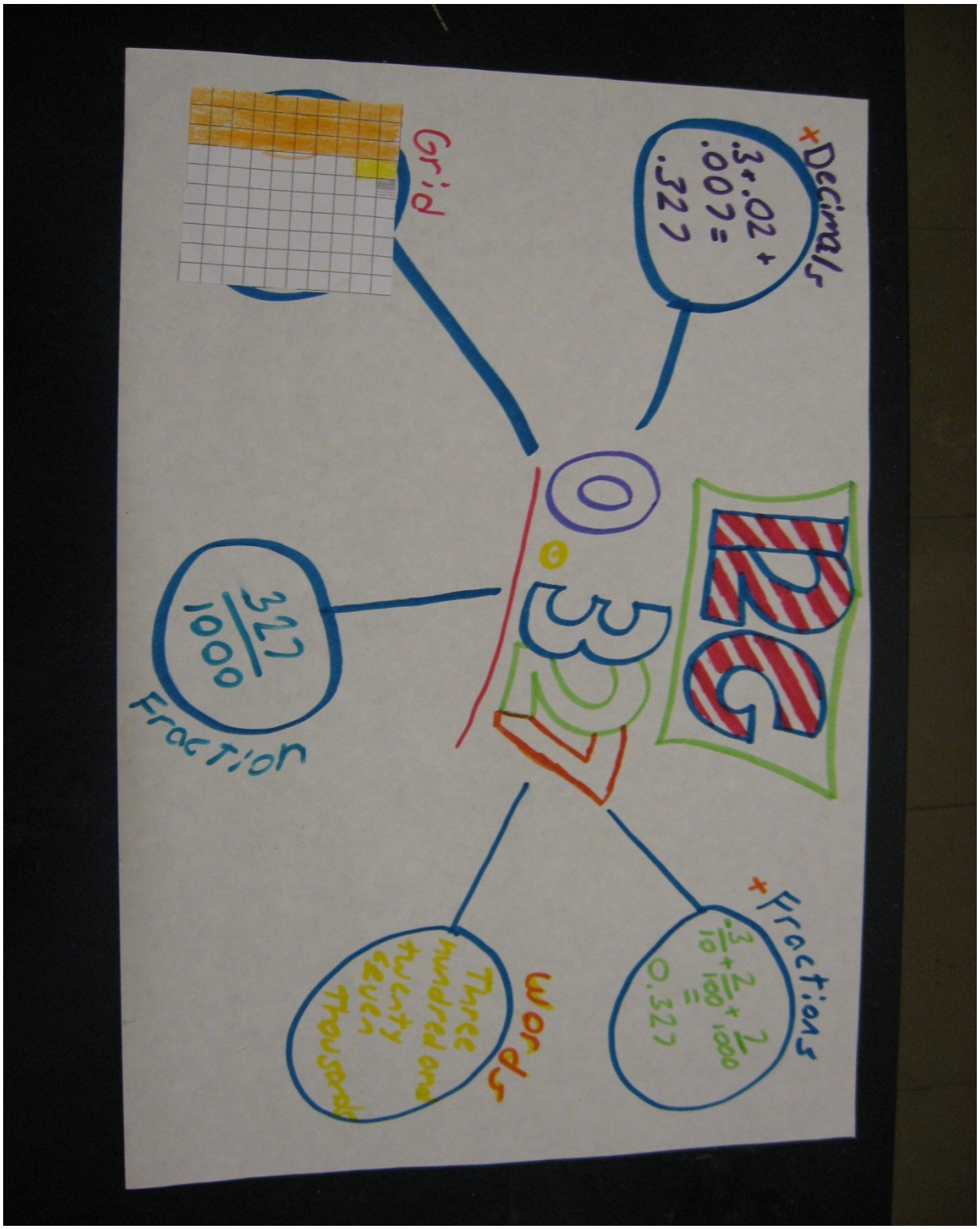
5. Explain that a good presentation clearly communicates not only the answer, but also how students reached the answer. Tell them to imagine that they are the teachers and they are to teach the rest of the students how to solve their problem.

6. Explain that a good audience listens politely to the presentation and asks questions if they don't understand something.

See Additional Notes for the Teacher for Lesson 12 to see examples of students' displays.

**Translations:**

- Symbolic to pictures to verbal
- Story problem to symbols to verbal
- Symbolic to symbolic to verbal


Grid
+Decimals
.3+.02+
.007=
.327
0.327
BG
3277/1000
Fraction
Words
Three hundred and twenty seven Thousandth
+Fractions
3/10+2/100+7/1000
=
0.327

Grid
decimal
0.327
words
three hundred twenty-seven thousands
Fraction
327/1000
Sum of decimals
.3
.02
+.007
.327
Sum of fractions
3/10 + 02/100 + 7/1000
007/1000 = 327/1000

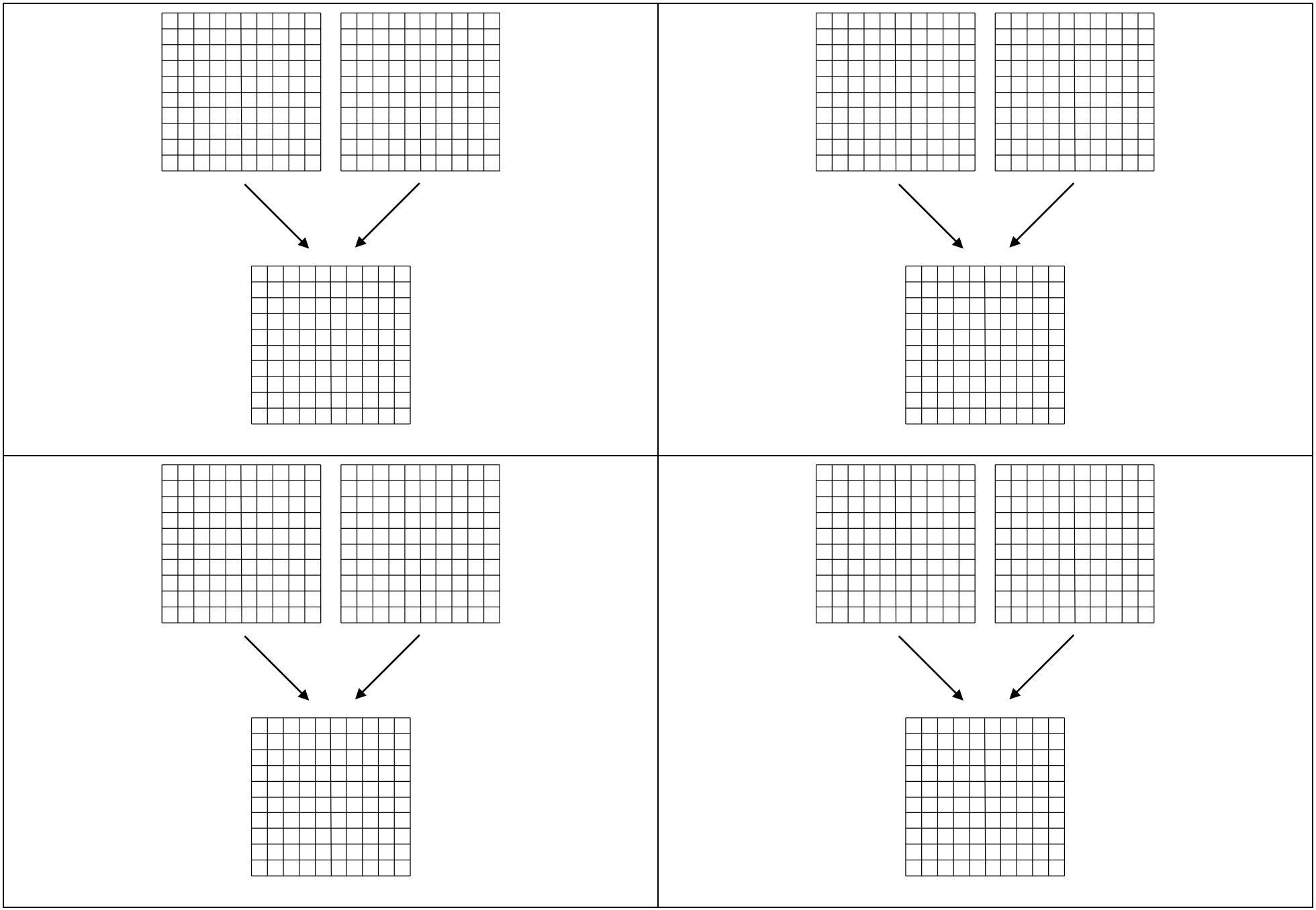
Order from Smallest-Largest.
Smallest
2
1
4
.5 largest
.005
.05
.062
.495
.5 also largest
Problem 12b
Use 10 x 10 grids to show how to order these decimals.
.005 .50 .05 .495 .062 .5

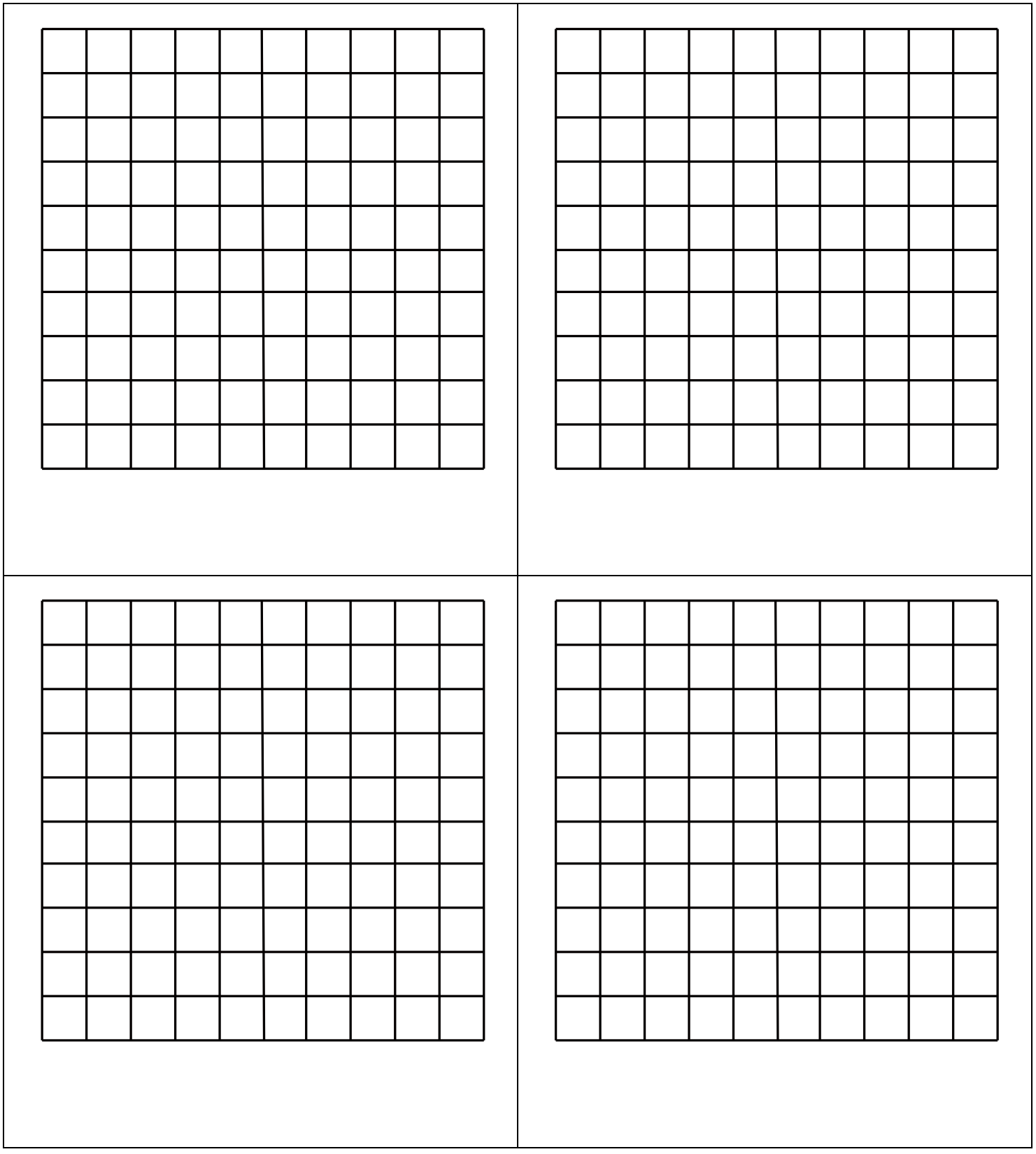
# Estimate

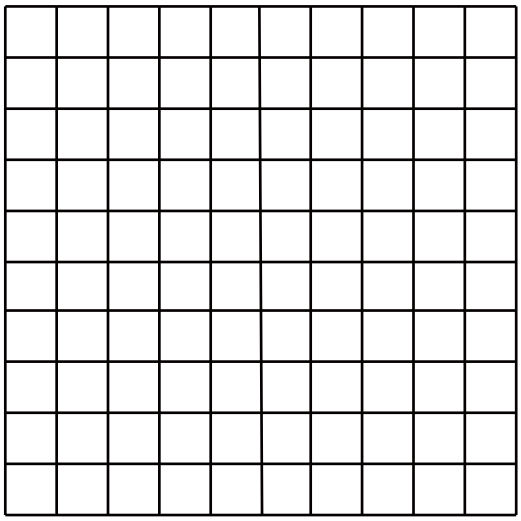
.85 + .39 =
Greater than 1 or less then 1?

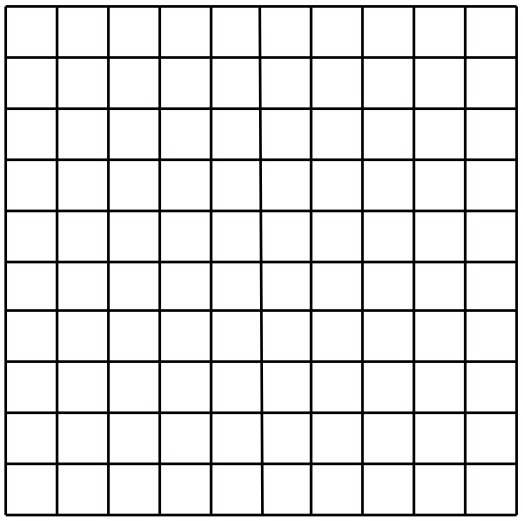
3.675 + 2.399 =
Greater then 6 or less than 6?

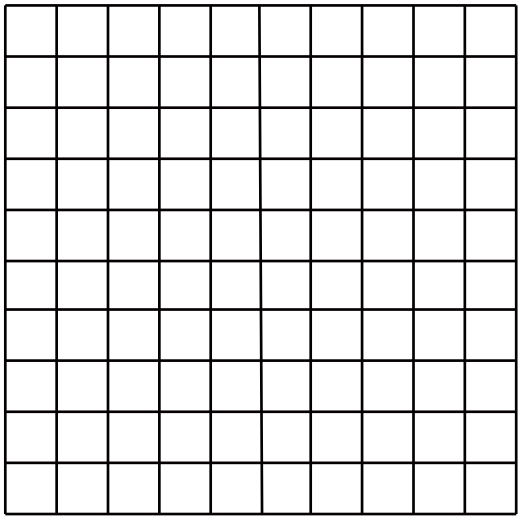
45.06 – 40.56 =
Greater than 5 or less than 5?

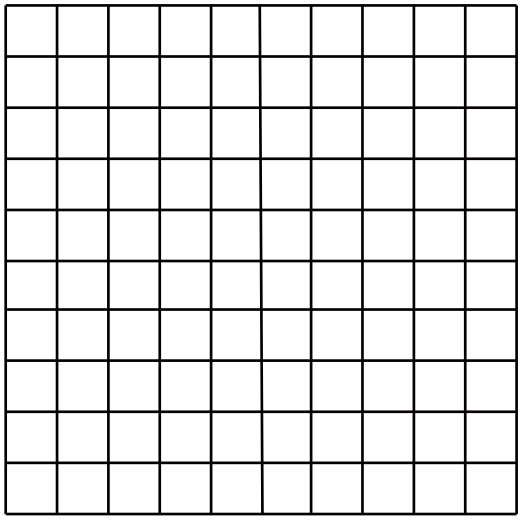


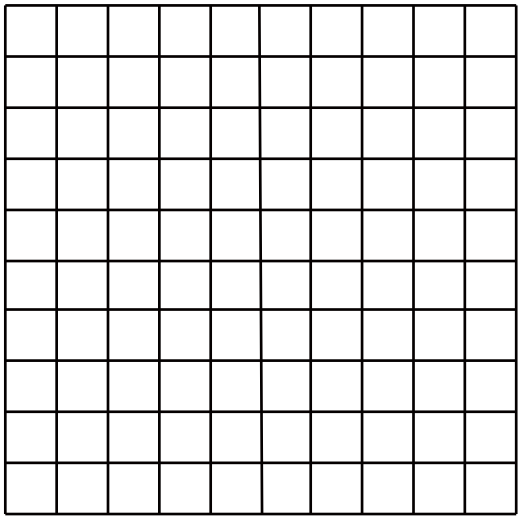


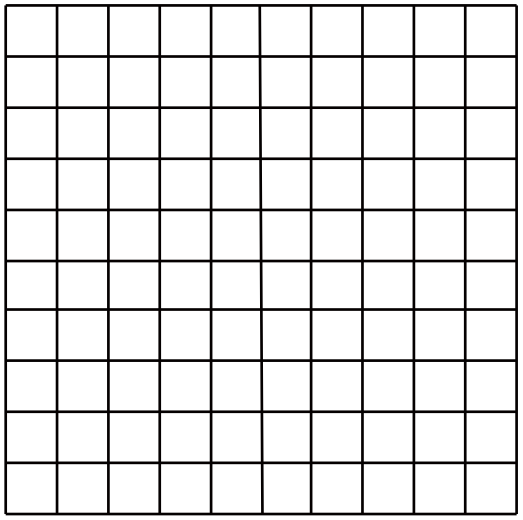












# Problems 12A – 12D

Copy and Cut out problems
Organize into envelopes for each group

| | |
|---|---|
| Problem 12A<br><br>Show how you can use decimal equivalents for each fraction to order the fractions from smallest to largest. Use your 10 x 10 grids in your solution.<br><br>$\frac{4}{5}$ $\frac{5}{10}$ $\frac{19}{20}$ $\frac{3}{4}$ | Problem 12B<br><br>Raven's tomato plant measured 15.035 cm when she planted it in May. By August it was 85.16 cm tall. How much did her plant grow from May to August?<br><br>• Draw a sketch showing the plant in May and in August.<br>• Estimate the plant growth. Did it grow more or less than 50 cm? More or less than 100 cm? Explain why.<br>• Show with symbols how to find the exact answer. |
| Problem 12C<br><br>Chee's sunflower plant grew 120.54 cm over the summer. When he planted it in May it measured 12.009 cm in height. How tall was his sunflower plant at the end of the summer?<br><br>• Draw a sketch showing the plant in May and at the end of the summer.<br>• Estimate the plant height. Was it more than 150 cm or less than 150 cm by the end of the summer? Explain why.<br>• Show with symbols how to find the exact answer. | Problem 12D<br><br>Use 10 x 10 grids to show how to order these decimals.<br><br>.005 .50 .05 .495 .062 .5 |

# Problems 12E – 12G

Copy and Cut out problems
Organize into envelopes for each group

Problem 12E

Estimate each answer before finding the exact answer. Explain your thinking.
Then show how to rewrite the problems so finding the exact answer is easy to do. Find the exact answers.

**4.7 + 3.94 + .034 =**
Estimate: Is the answer greater than 10 or less than 10? Is it greater than 400 or less than 400? Is it greater than 6 or less than 6?

**16.75 – 8.974=**
Estimate: Is it greater than 25 or less than 25? Is the answer greater than 10 or less than 10? Is it greater than 6 or less than 6?

**53 – 4.95 =**
Estimate: Is the answer greater than 10 or less than 10? Is it greater than 25 or less than 25? Is it greater than 50 or less than 50?

Problem 12F

Imagine a 10 x 10 grid. Picture each decimal below given in words on this grid. Describe what you see in your mind.

Now write each decimal in symbols.

- 12-tenths
- 105-hundredths
- 33-tenths
- 1002-thousandths

Problem 12 G

Show this number in multiple ways (fraction, decimal, grid, words, as a sum of two or more decimals or fractions…)

0.327

Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) To improve the lesson I suggest: